

Sparse Computable Erasures

and Least-Degree Obstructions

Low counterexamples, finite meets, and descending chains
in effective-dense equivalence classes

Research draft prepared for Vladimir Reshetnikov

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Abstract

We give a negative answer to the effective-dense least-representative question C2 in the PROVEIT research programme, in both its uniform and nonuniform versions. If a binary set G is 1-generic and computes no function eventually different from every computable function, neither equivalence class of G has an element of least ordinary Turing degree. Every 2-generic satisfies the hypothesis. A classical highness characterization also supplies all non-high 1-generics, including low Δ_2^0 examples.

The key step turns a density-zero omission set into a bounded choice function. An infinitely often computable agreement yields a decidable sparse mask that escapes that omission set. Genericity then prevents recovery of an effective-dense description from the erased oracle. For every unbounded computable order b , the mask can satisfy $|T \cap [0, N]| \leq b(N)$ at every prefix. We also identify the exact lattice of degrees arising from computable masks, prove a uniform countable simultaneous-avoidance theorem, and construct a strictly descending chain inside one uniform class, all within the same budget, with no lower bound belonging to the full nonuniform class. Every member of the chain can be low.

The main erasure argument and the 2-generic route are proved directly. The low-example refinement explicitly uses the classical Kjos-Hanssen–Merkle–Stephan characterization, rather than presenting it as new. The article records its source audit, formalization obligations, and further questions.

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[write] **Research report 14 of the CoarseDegrees programme** (batch 37, manuscript 11). Text marked [write] was added when it was filed; Sections 1.4–1.7 give the notation against the plan and the other round-2 reports, the provenance, the formal status and the plan targets addressed. The delivered `REPOSITORY_UPDATE.md` is filed verbatim and has not been applied. Nothing in this report is formalized. It is AI-assisted and unrefereed.

1 The question and the results

1.1 A specific target, not a repetition of the coarse case

The PROVEIT research plan distinguishes two problems that are easy to confuse. Its earlier coarse least-representative question, C1, is retired; its effective-dense counterpart, C2, is retained with a proposed genericity argument and an explicitly identified gap. The present paper addresses C2. The repository snapshot used here is commit `1a1396d4d3a2ac6812692df520517d86ba3a4785`; the relevant path and source locations are recorded in Appendix A.[1]

An effective-dense description may withhold an answer on a density-zero set, but may never give a wrong numerical answer. Withholding is represented by an explicit symbol $\perp$, not by a computation that fails to halt. Thus arbitrary density-zero modifications do not automatically preserve the associated description problem. Our modifications have a *computable* support, which makes two correct, uniform description transformations possible.

For $r \in \{\text{ued}, \text{ned}\}$, let $[f]_r$ denote the r -equivalence class of f , and put

$$\text{Spec}_T([f]_r) = \{\deg_T(g) : g \in [f]_r\}.$$

The target is the assertion

$$(\forall f \in \mathbb{N}^{\mathbb{N}})(\exists g \in [f]_r)(\forall h \in [f]_r) g \leq_T h. \quad (1)$$

This is the effective-dense instance of Gerdes’s Question 7. His question also asks more generally about the spectra of ordinary Turing degrees of representatives.[2, Question 7] The word “ordinary” matters: members of $[f]_r$ have one and the same asymptotic degree, but can have quite different Turing degrees.

1.2 A hypothesis separating prediction from computational strength

Say that an oracle G has the *computable-agreement property* if

$$\text{CA}(G) : \quad (\forall f \in \mathbb{N}^{\mathbb{N}})[f \leq_T G \implies (\exists h \text{ total computable})(\exists^\infty n) f(n) = h(n)]. \quad (2)$$

This is simply the assertion that G computes no function eventually different from every total computable function. The notation is local to this paper; the property itself is classical. Its highness-theoretic interpretation appears in Section 5.[4, 5]

Theorem 1.1 (Least-degree obstruction). *Let $G \in 2^{\mathbb{N}}$ be 1-generic and satisfy $\text{CA}(G)$. Neither $\text{Spec}_T([G]_{\text{ued}})$ nor $\text{Spec}_T([G]_{\text{ned}})$ has a least element. The assertion remains true when the representatives are restricted to binary sets. Such G can be chosen low and computable from $\emptyset'$.*

Here “low” means $G' \equiv_T \emptyset'$. The hypotheses hold for every 2-generic and, by a classical theorem cited explicitly below, every non-high 1-generic. No assertion for *all* 1-generics is needed to refute (1).

For a computable set T , write

$$G_T(n) = \begin{cases} 0, & n \in T, \\ G(n), & n \notin T. \end{cases} \quad (3)$$

The mask T is a set of coordinates whose information is discarded. The output G_T is still binary. If T has density zero, then $G_T \equiv_{\text{ued}} G$.

[write] *Not the plan’s* G_T . The research plan’s line of attack on C2 (`turing_degrees_unified.tex`, line 570) writes G_T for the function equal to 2 on T and to G elsewhere. Here G_T is 0

on T and stays binary. For computable null T both are uniformly effective-dense equivalent to G and Turing equivalent to $G \upharpoonright (\mathbb{N} \setminus T)$, so every degree statement below holds for either reading; but the two are different functions, and only the binary one is a set representative. See Section 1.4.

Theorem 1.2 (Arbitrarily sparse local avoidance). *Let G satisfy the hypotheses of Theorem 1.1. Suppose $A \leq_T G$ computes an effective-dense description of G , and let $b : \mathbb{N} \rightarrow \mathbb{N}$ be computable, nondecreasing, and unbounded. There is an infinite computable density-zero set T such that*

$$|T \cap [0, N]| \leq b(N) \quad (N \in \mathbb{N}), \quad G_T \equiv_{\text{ued}} G, \quad G_T <_T G, \quad A \not\leq_T G_T.$$

In particular, $|(G \triangle G_T) \cap [0, N]| \leq b(N)$ for every N .

The candidate A need not itself be an equivalent representative: computing one effective-dense description is enough. The theorem is an existence statement for a computable mask, not a uniform algorithm which extracts its index from arbitrary oracle data.

Theorem 1.3 (Packed descending chains). *For the same G and b , there are computable density-zero masks $C_0 \subseteq C_1 \subseteq \dots$ such that, with $G_i = G_{C_i}$,*

$$G_0 >_T G_1 >_T G_2 >_T \dots, \quad G_i \equiv_{\text{ued}} G, \quad |(G \triangle G_i) \cap [0, N]| \leq b(N).$$

There is no $g \in [G]_{\text{ned}}$ with $g \leq_T G_i$ for every i . If G is low, every G_i is low.

A countable descending chain need not be cointial in the entire class. Nor does the last assertion deny the existence of lower bounds outside that class: computable sets are always ordinary Turing lower bounds. Both qualifications are essential.

1.3 What is contributed and what is classical

The contribution developed here is the passage from computable agreement and generic non-prediction to least-degree obstructions for effective-dense representatives, together with its quantitative and countable refinements. The no-eventually-different-function fact about 2-generics and the high-or-DNR characterization are classical.[5, 4] We include a direct proof of the former and invoke the latter only for the low refinement. The finite-coordinate arguments and elementary density estimates are proved rather than treated as new background theorems.

The audit located the explicit least-representative question in Gerdes’s 2025 preprint and the distinct quasiminimality statements in Astor–Hirschfeldt–Jockusch. Quasiminimality concerns a particular embedding of ordinary Turing degrees into asymptotic degrees; it is not the same statement as nonexistence of a least ordinary degree *within* a fixed class.[3, Theorem 5.9] No claim of first publication or exhaustive literature coverage is made.

1.4 [write] Notation against the research plan and the other round-2 reports

The effective-dense notions here are those of the research plan’s target C2 [1]: $\equiv_{\text{ned}}$ and $\equiv_{\text{ued}}$, with omissions written as an explicit $\perp$ (the strong-domain convention of [3]). The density-zero sets $\mathcal{Z}_0$ are the sets the Lean library of the project calls `DensityZero`, and “1-generic” is its predicate `OneGeneric`. No symbol was renamed and no normalization changed when this report was written. The other round-2 reports (11–13) are about *coarse* descriptions, which may err on a null set; several of their letters mean something else here.

Symbol	Meaning here	Do not read it as
G_T	0 on T , G elsewhere; binary	the plan's G_T (plan, line 570), which is 2 on T
$\text{Spec}_T([f]_r)$	Turing spectrum of an <i>effective-dense</i> class, $r \in \{\text{ued}, \text{ned}\}$	the coarse spectra of the synthesis and of reports 11 and 13
T, C_i	computable null <i>masks</i> ; C_i is the i th mask of the chain (13)	the dyadic columns C_i of the synthesis and report 13, or report 13's tails T_k
I_n, J_n	blocks $[m_n, 2m_n)$ of positions; dyadic blocks $[2^n, 2^{n+1})$ in Lemma 7.1	a block <i>code</i> such as the plan's J , or report 13's $I_n = [2^n - 1, 2^{n+1} - 1)$
P_T	the projection $X \mapsto X_T$	report 13's protected regions P_k , or report 11's presentation P
$r_k(n)$	block fractions in Lemma 7.1	report 11's column map $r_k(t)$
$\text{CA}(G)$	G computes no function eventually different from all computable ones; local notation	a notion defined elsewhere in the programme
“low”	$G' \equiv_T \emptyset'$	report 13's “low over $\emptyset'$ ”, i.e. $(Z \oplus \emptyset')' \equiv_T \emptyset''$

1.5 [write] Provenance of this report

The report is built from one manuscript: batch 37 manuscript 11, the archive `proveit_effective_dense_research` (inner directory `proveit_effective_dense/`, main file `article.tex`, 22-page PDF), which arrived in commit `37e61c1fd`. Its pin is the commit printed above, `1a1396d4d3a2ac6812692df520517d86ba3a4785` (the batch-36 placement commit), and it was placed as report 14 in commit `0e53d1034`. Between the pin and the placement, `Computability/TuringDegrees/` changed only by the addition of the round-2 reports 11–14, so the plan blob and the line range quoted in Appendix A are those of the current tree.

When the report was written, every label received the prefix `cr14:`, the title page received the page-anchor guard that removes a duplicate PDF destination of the delivered source, and the note below the contents, these four subsections, and [write] notes after (3), in Section 9.2 and in Section 11.3 were added. No statement of the manuscript was changed. Every numbered statement, equation and section keeps its number in the delivered PDF; that PDF is not shipped, and `article.pdf` is a build of this text.

The delivered files are shipped under new paths: `checks/check_finite.py` is `code/check_finite.py`, `checks/results.json` and `checks/run.txt` are `data/results.json` and `data/run.txt` (the two are byte-identical), and `build.sh` is `code/build.sh`; `PROOF_AUDIT.md`, `SOURCES.md` and `REPOSITORY_UPDATE.md` keep their names. The delivered checksum file is not shipped. `REPOSITORY_UPDATE.md` proposes new wording for the plan's C2 entry. It is filed verbatim as a record of the delivery and has *not* been applied: the plan, its README and the synthesis are unchanged.

1.6 [write] Formal status and the Lean library

No statement of this report is formalized, in Lean or in Rocq; that it sits in the research programme of the Lean/Rocq project `Computability/TuringDegrees` gives it no formal status. The project's Lean library `CoarseDegrees` (`Computability/TuringDegrees/Lean/CoarseDegrees`) formalizes coarse, not effective-dense, descriptions: it has no $\perp$ -valued descriptions, no $\equiv_{\text{ued}}$ or $\equiv_{\text{ned}}$, no eventually different or DNR functions, no highness and no jump operator. The report relies on none of its eight admitted statements. What it shares with the library is basic: the predicate `OneGeneric` (`Sets.lean`, line 136) has the definition used here, and the existence of 1-generic sets is proved there as `exists_oneGeneric` (`Generic.lean`, line 122), without the bound $G \leq_T \emptyset'$ of Proposition 5.5. The external input of the low refinement, Theorem 5.2, has

no Lean statement. Question 10.12 and Section 11.2 are a plan: no Lean or Rocq code was delivered or is shipped.

1.7 [write] Programme targets addressed, and what stays open

Line numbers refer to `Computability/TuringDegrees/Research/CoarseDegrees/research-plan/turing_degrees_unified.tex` (“plan”), its `README.md`, and `.../research-synthesis/Turing_Degrees_Synthesis.tex` (“synthesis”), all identical at the pin and in the current tree.

- *C2* (plan:563–572; plan README:20–22; synthesis:969, “None of the reports bears on the effective-dense question in general”; plan:1647 lists “C2 along the line of attack” among the first choices for fresh work). Theorem 6.2 refutes the universal C2 statement, in both the uniform and the nonuniform version, for every 1-generic with $\text{CA}(G)$: all 2-generics directly, all non-high 1-generics through the external Theorem 5.2, and low Δ_2^0 examples. The plan’s line of attack (plan:570) is followed in its first two steps (Lemma 3.1 is step (i) with 0 in place of 2; Corollary 3.3 is step (ii)), and its step (iii), “the gap”, is *bypassed, not proved*: the hypothesis $\text{CA}(G)$ and Theorem 4.4 replace the forcing argument the plan asks for. For 1-generics without CA , which are exactly the high ones by Theorem 5.2 and Lemma 5.3, step (iii) is still open (Question 10.1). So the C2 question is answered negatively (unrefereed), and the gap in the plan’s sketch is re-scoped to high 1-generics, not closed. The “weaker, still informative result” suggested at plan:572 is exceeded for these G ; the positive test on the guarded c.e. set suggested there is not attempted; the literature audit suggested there was done for Astor–Hirschfeldt–Jockusch and Gerdes, and is bounded (Appendix A).
- *C3* (plan:574): not solved; Question 10.11 notes a possible connection.
- *Minimal elements and coinitality* for effective-dense classes are open (Questions 10.4 and 10.5); the synthesis items for coarse spectra (synthesis:970, 973) and C4–C8 are not addressed.
- The delivered `REPOSITORY_UPDATE.md` proposes the corresponding change of the plan’s C2 status. It is not applied here; amending the plan and the synthesis is separate programme work.

2 Definitions and conventions

2.1 Descriptions and their two reducibilities

We identify a set with its characteristic function. All ordinary functions under discussion are total unless explicitly called partial. A function oracle can equivalently be coded by its graph, so ordinary Turing reducibility between sets and functions is unambiguous.

For $S \subseteq \mathbb{N}$ and $N \geq 1$, put

$$\rho_N(S) = \frac{|S \cap [0, N]|}{N}, \quad \mathcal{Z}_0 = \{S : \lim_{N \rightarrow \infty} \rho_N(S) = 0\}.$$

We write $S \subseteq^* T$ when $S \setminus T$ is finite. Finite unions of members of $\mathcal{Z}_0$ belong to $\mathcal{Z}_0$, since prefix counts are subadditive.

Definition 2.1. An *effective-dense description* of $f \in \mathbb{N}^{\mathbb{N}}$ is a total function $d : \mathbb{N} \rightarrow \mathbb{N} \cup \{\perp\}$ satisfying

$$d(n) \in \{f(n), \perp\} \quad (n \in \mathbb{N}), \quad S_d := \{n : d(n) = \perp\} \in \mathcal{Z}_0.$$

Let $\text{ED}(f)$ be the collection of all such descriptions.

The explicit output $\perp$ can be coded by a distinguished natural number, with numerical answers shifted by one. This coding does not change degrees. The definition agrees with the strong-domain convention for effective dense computability.[3, Definitions 1.4 and 3.1] In particular, f itself is a member of $\text{ED}(f)$, and if $d \leq_T A$ then $S_d \leq_T A$.

Definition 2.2. For $f, g \in \mathbb{N}^{\mathbb{N}}$,

$$f \leq_{\text{ned}} g \iff (\forall d \in \text{ED}(g))(\exists e \in \text{ED}(f)) e \leq_T d.$$

We have $f \leq_{\text{ued}} g$ if there is one oracle Turing functional Φ such that Φ^d is total and belongs to $\text{ED}(f)$ for every $d \in \text{ED}(g)$. Equivalence means reduction in both directions.

The uniform relation implies the nonuniform relation. Reflexivity is witnessed by copying a description. For transitivity, compose the two functional transformations in the uniform case; in the nonuniform case, choose a description at the intermediate step and use transitivity of $\leq_T$. Thus the indicated equivalence classes are well-defined.

A key implication used repeatedly is

$$G \leq_{\text{ned}} g \implies (\exists d \in \text{ED}(G)) d \leq_T g. \quad (4)$$

It follows by using g itself as the input description of g . It does not say that an arbitrary description of G , recoded as a total numerical function, must be an equivalent representative.

2.2 Finite oracle computations and genericity

Let $2^{<\mathbb{N}}$ be the finite binary strings. The relation $\sigma \preceq X$ means that σ is an initial segment of X ; $\tau \succeq \sigma$ means extension. A computation $\Phi^\sigma(n) \downarrow = v$ is witnessed by a finite run using only coordinates below $|\sigma|$. It persists under extensions. Finite convergence is computably enumerable, and convergence within a specified number of steps is decidable.

Definition 2.3. A binary set G is *1-generic* if, for every computably enumerable $W \subseteq 2^{<\mathbb{N}}$, either some $\sigma \preceq G$ belongs to W , or some $\sigma \preceq G$ has no extension in W . It is *2-generic* if the same condition holds for every Σ_2^0 set of strings.

Avoiding a set of strings always means the stronger, cylinder-wise alternative in this definition, not merely having no prefix in the set. The latter alone is insufficient for the arguments below. A 2-generic is 1-generic. If a c.e. set of strings is dense above a prefix of a 1-generic, that generic must meet it.

For fixed computable T , let $P_T(X) = X_T$ as in (3); for a finite string, $P_T(\sigma)$ is the same-length string obtained by zeroing its T coordinates. The map is computable, preserves extension, and ignores the actual input bits on T .

2.3 Least, minimal, and cointial

For a class $\mathcal{C}$ of oracles, $g \in \mathcal{C}$ has least degree in $\mathcal{C}$ if $g \leq_T h$ for every $h \in \mathcal{C}$. It has minimal degree in $\mathcal{C}$ if no $h \in \mathcal{C}$ satisfies $h <_T g$. Least implies minimal, but the converse need not hold. A subfamily $\mathcal{A} \subseteq \mathcal{C}$ is *cointial* if every member of $\mathcal{C}$ computes some member of $\mathcal{A}$. We shall exclude finite cointial families and certain uniformly presented countable ones. We do not exclude arbitrary countable cointial families.

3 Computable erasures and the generic mask lattice

3.1 Two transformations that never give a wrong answer

Lemma 3.1 (Computable-null erasure). *For every binary X and computable $T \in \mathcal{Z}_0$, we have $X_T \equiv_{\text{ued}} X$ and $X_T \leq_T X$.*

Proof. Given $d \in \text{ED}(X)$, define

$$F_T(d)(n) = \begin{cases} 0, & n \in T, \\ d(n), & n \notin T. \end{cases}$$

This is an effective-dense description of X_T : it is correct on T , correct or $\perp$ elsewhere, and has omission set $S_d \setminus T$. Given $e \in \text{ED}(X_T)$, define

$$H_T(e)(n) = \begin{cases} \perp, & n \in T, \\ e(n), & n \notin T. \end{cases}$$

It describes X , with omission set $T \cup S_e$. Both sets of omissions are null. The two transformations are single total oracle procedures with the decidable mask hard-coded; no search for a density-convergence rate is used. Finally, an oracle for X computes X_T directly. $\square$

3.2 A deleted coordinate cannot be predicted infinitely often

Lemma 3.2 (Generic nonprediction). *Let G be 1-generic and let T be computable. Suppose a partial G_T -computable function ψ has the property that, whenever $n \in T$ and $\psi(n) \downarrow$, its value is $G(n)$. Then $T \cap \text{dom } \psi$ is finite.*

Proof. Fix an index e with $\psi = \Phi_e^{G_T}$. Let W be the c.e. set of strings σ for which there are $n \in T$ below $|\sigma|$ and $b \in \{0, 1\}$ such that

$$\Phi_e^{P_T(\sigma)}(n) \downarrow = b, \quad \sigma(n) = 1 - b.$$

No prefix of G is in W , since such a finite computation would persist on G_T and give an incorrect prediction. By 1-genericity, choose $\sigma_0 \preceq G$ with no extension in W .

Suppose $n \in T$, $n \geq |\sigma_0|$, and $\Phi_e^{G_T}(n) \downarrow = b$. Take a sufficiently long prefix τ of G to witness this convergence and include coordinate n . Replace the n th bit of τ by $1 - b$, obtaining $\hat{\tau}$. This still extends σ_0 , and $P_T(\hat{\tau}) = P_T(\tau)$ because $n \in T$. Thus $\hat{\tau} \in W$, a contradiction. All convergent inputs in T are therefore below $|\sigma_0|$. $\square$

Corollary 3.3 (Forced omissions). *If G is 1-generic, T is computable, and $d \in \text{ED}(G)$ satisfies $d \leq_T G_T$, then $T \subseteq^* S_d$.*

Proof. On input n , compute $d(n)$ and return its value if numerical; otherwise run forever. This gives a partial G_T -computable predictor that is always correct. Its convergent inputs in T are exactly $T \setminus S_d$. Apply Lemma 3.2. $\square$

3.3 An exact order calculation

Theorem 3.4 (Mask order). *For a 1-generic G and computable $C, D \subseteq \mathbb{N}$,*

$$G_C \leq_T G_D \iff D \subseteq^* C. \tag{5}$$

Consequently $G_C \equiv_T G_D$ exactly when $C \triangle D$ is finite.

Proof. If $D \setminus C$ is finite, compute G_C from G_D by outputting zero on C , copying G_D off $C \cup D$, and hard-coding the finitely many correct bits on $D \setminus C$.

Conversely, suppose $G_C \leq_T G_D$ and $D \setminus C$ is infinite. On that infinite computable set, G_C supplies the original bits of G . It would therefore be a partial G_D -computable predictor correct on an infinite subset of D , contrary to Lemma 3.2. $\square$

In particular, every infinite computable mask strictly lowers the degree of a 1-generic. If $C \subseteq D$ and $D \setminus C$ is infinite, then $G_D <_T G_C$. This does not require G_C itself to be generic.

Theorem 3.5 (Finite joins and actual greatest lower bounds). *For computable C, D and a 1-generic G ,*

$$G_C \oplus G_D \equiv_T G_{C \cap D}, \quad \text{Low}(G_C) \cap \text{Low}(G_D) = \text{Low}(G_{C \cup D}),$$

where $\text{Low}(X) = \{A : A \leq_T X\}$. Thus the second identity gives a greatest lower bound in the full Turing degrees, not merely inside the mask family.

Proof. The join identity is direct. Off $C \cap D$, at least one oracle retains the original bit; decidability of C, D tells us which one to use. Conversely, $G_{C \cap D}$ computes both masked oracles by further zeroing. Also $G_{C \cup D}$ computes from either, so one inclusion in the lower-cone identity is immediate.

For the other, fix a total function $A = \Phi^{G_C} = \Psi^{G_D}$. Let W be the c.e. set of strings σ witnessing, at some input n ,

$$\Phi^{P_C(\sigma)}(n) \downarrow \neq \Psi^{P_D(\sigma)}(n) \downarrow.$$

No prefix of G lies in W . Choose $\sigma_0 \preceq G$ such that no extension is in W . Put $H = G_{C \cup D}$.

To compute $A(n)$ from H , search for a finite string τ extending σ_0 which agrees with H at all coordinates outside $C \cup D$ below $|\tau|$, and for which $\Phi^{P_C(\tau)}(n) \downarrow = v$. The compatibility test is decidable in H ; the search halts because a sufficiently long actual prefix of G is a witness.

We claim that every value found equals $A(n)$. Take a long actual prefix $\eta \preceq G$, extending σ_0 , which witnesses $\Psi^{P_D(\eta)}(n) \downarrow = A(n)$. The private coordinates used by the first oracle are in $D \setminus C$; those used by the second are in $C \setminus D$. They are disjoint. On the shared coordinates outside $C \cup D$, the two strings agree by the compatibility test. On $C \cap D$, both projected oracles are zero. We may therefore amalgamate the two finite computations into one string $\nu \succeq \sigma_0$, preserving the first computation from τ and the second from η . If $v \neq A(n)$, this gives $\nu \in W$, a contradiction. The search computes A from H . $\square$

It follows that

$$(\text{Comp} \cap \mathcal{Z}_0)/\text{Fin} \longrightarrow \text{Spec}_T([G]_{\text{ued}}), \quad [C] \longmapsto \deg_T(G_C),$$

is an order-reversing injection with union corresponding to meet and intersection to join. This is a precise lattice structure *inside one* uniform effective-dense class. It has no least member: any computable null C can be enlarged by an infinite computable null subset of its complement. For example, find successively a point outside C beyond 2^n and the previous chosen point. Theorem 3.4 then supplies a strictly smaller member. This last observation by itself does not rule out a least representative lying outside the mask family; that is the purpose of the next sections.

4 Escaping a null set with a computable sparse selector

4.1 Packing prescribed capacities into every-prefix budgets

Lemma 4.1 (Budgeted blocks). *Let $b : \mathbb{N} \rightarrow \mathbb{N}$ be computable, nondecreasing, and unbounded, and let (a_n) be a computable sequence of positive integers. Put $Q_n = \sum_{j \leq n} a_j$. There is a computable sequence (m_n) such that*

$$m_n \geq 4^{n+1}, \quad m_{n+1} \geq 4m_n, \quad m_n \geq 2n + 3, \quad (6)$$

$$m_n \geq (n+1)Q_n, \quad b(m_n) \geq Q_n. \quad (7)$$

Let $I_n = [m_n, 2m_n)$. Any set T contained in $\bigcup_n I_n$ with $|T \cap I_n| \leq a_n$ satisfies

$$|T \cap [0, N]| \leq b(N) \quad (N \in \mathbb{N}), \quad T \in \mathcal{Z}_0.$$

No computability of T is needed for this last conclusion.

Proof. Find m_n successively by an ordinary search satisfying the finitely many lower bounds and $b(m_n) \geq Q_n$. Unboundedness of b makes each search halt. The intervals are disjoint by (6). For $N < m_0$ the count is zero. If $m_n \leq N < m_{n+1}$, then

$$|T \cap [0, N]| \leq Q_n \leq b(m_n) \leq b(N).$$

Moreover, for $N \geq 1$ in the same range,

$$\rho_N(T) \leq Q_n/m_n \leq 1/(n+1).$$

As N tends to infinity, so does n , proving density zero. Notice that this estimate covers prefixes *inside* a block as well as endpoints. $\square$

4.2 One missed point per interval

Definition 4.2. For any sequence of blocks as in Lemma 4.1 and any set S , define

$$f_S(n) = \begin{cases} \min\{k < m_n : m_n + k \notin S\}, & I_n \not\subseteq S, \\ 0, & I_n \subseteq S. \end{cases} \quad (8)$$

For a total computable $h : \mathbb{N} \rightarrow \mathbb{N}$, define its selector

$$T_h = \{m_n + (h(n) \bmod m_n) : n \in \mathbb{N}\}. \quad (9)$$

The function f_S is total and computable from S . It requires only a finite search on each input, even if m_n is very large. The set T_h is computable: find the unique block which could contain the proposed input, if one exists, and compute the one selected point in that block. It is not merely c.e. The total running time of $h(n)$ need not have been predicted when the interval endpoints were chosen.

Lemma 4.3 (Null covers imply eventual difference). *Suppose $S \in \mathcal{Z}_0$. For all sufficiently large n , $I_n \not\subseteq S$. If $T_h \subseteq^* S$ for every total computable h , then f_S is eventually different from every total computable function.*

Proof. If infinitely many I_n were contained in S , then at the corresponding endpoints $\rho_{2m_n}(S) \geq 1/2$, contradicting density zero. Fix computable h . For all sufficiently large n , the chosen point of T_h belongs to S , whereas $m_n + f_S(n)$ is outside S . If $f_S(n) = h(n)$, then $h(n) < m_n$ and the two points coincide, a contradiction. Thus $f_S(n) \neq h(n)$ eventually. $\square$

Theorem 4.4 (Budgeted sparse escape). *If $\text{CA}(G)$, $S \leq_T G$, and $S \in \mathcal{Z}_0$, then for every unbounded computable order b there is an infinite computable $T \in \mathcal{Z}_0$ with*

$$|T \cap [0, N]| \leq b(N) \quad (N \in \mathbb{N}), \quad |T \setminus S| = \infty. \quad (10)$$

Proof. Use Lemma 4.1 with $a_n = 1$. The total function f_S of (8) is computable from G . By $\text{CA}(G)$, some total computable h agrees with it infinitely often. On all sufficiently large agreement inputs, the corresponding point in T_h is a genuine point outside S . Thus $T = T_h$ has infinitely many points outside S . It has exactly one point in each block, so it is infinite, is computable, and satisfies the budget and density conclusions of Lemma 4.1. $\square$

Remark 4.5 (What happened to the timing gap?). The argument never attempts to enumerate a sparse set of potential witnesses and then infer that the enumeration has a computable membership test. It first uses $\text{CA}(G)$ to obtain a *total computable function* h , and then defines exactly one point in each computably located interval. Unbounded convergence times cause no difficulty for a membership algorithm which is permitted to wait for a guaranteed total computation. This is the specific replacement for the delayed-witness construction discussed in C2.

A useful independent consequence is that any oracle computing a null set which almost contains every computable null set must compute an eventually different function. Indeed all the selectors in (9) are computable and null. The choice function is even bounded by the computable function m_n . This observation explains why the existence of abstract null upper bounds does not settle an effective question about them.

5 Classical sources of computable agreement

5.1 A direct proof for every 2-generic

The following fact is classical; it is consistent with the formulation in terms of weak meager engulfing recorded by Brendle–Brooke–Taylor–Ng–Nies. We supply the finite-extension proof needed here.[5, Theorem 4.2 and Section 4, item (5)]

Theorem 5.1. *Every 2-generic G satisfies $\text{CA}(G)$.*

Proof. Fix a functional Γ with $f = \Gamma^G$ total. Consider

$$B_\Gamma = \{\sigma : (\exists n)(\forall \tau \succeq \sigma)\Gamma^\tau(n) \uparrow\}.$$

Here divergence of a finite-oracle computation means that no finite-time convergence witness exists. The displayed condition has the form $\exists n \forall (\tau, s)$ with a decidable matrix, so B_Γ is Σ_2^0 . No prefix of G is in it, because Γ^G is total. There is therefore $\sigma_0 \preceq G$ such that no extension of σ_0 is in B_Γ . Equivalently,

$$(\forall \tau \succeq \sigma_0)(\forall n)(\exists \nu \succeq \tau) \Gamma^\nu(n) \downarrow. \quad (11)$$

Choose a computable sequence (τ_n) listing every finite extension of σ_0 infinitely often. On input n , search for an extension $\nu \succeq \tau_n$ with $\Gamma^\nu(n) \downarrow$, and set $h(n)$ to the value found. Equation (11) makes h total and computable. The finite string σ_0 is hard-coded; it need not be computably selected from an index for G .

For each k , the set

$$U_k = \{\nu : (\exists n \geq k) \Gamma^\nu(n) \downarrow = h(n)\}$$

is c.e. and dense above σ_0 . To see density, take any extension τ of σ_0 , choose $n \geq k$ with $\tau_n = \tau$, and use the extension found while defining $h(n)$. Since G is also 1-generic, it meets U_k . Thus $f(n) = h(n)$ for some $n \geq k$, for every k . $\square$

This theorem, together with the preceding sections and Section 6, gives a proof of the negative least-degree answer without importing a highness classification or a basis theorem.

5.2 Non-high 1-generics and low examples

A total function $d : \mathbb{N} \rightarrow \mathbb{N}$ is *diagonally nonrecursive*, or DNR, if $d(e) \neq \varphi_e(e)$ whenever the latter is defined. An oracle has DNR degree if it computes such a function. It is *high* if its jump computes $\emptyset''$.

Theorem 5.2 (Classical characterization; external input). *An oracle computes a function eventually different from every total computable function if and only if it is high or has DNR degree.*

This is Kjos-Hanssen–Merkle–Stephan, Theorem 5.1.[4, Theorem 5.1] We use its stated equivalence, not any of its Kolmogorov-complexity characterizations. The theorem is not a result of the present paper.

Lemma 5.3 (A standard genericity consequence). *No 1-generic computes a DNR function.*

Proof. Suppose $d = \Gamma^G$ is total and DNR. The set

$$W = \{\sigma : (\exists e) \Gamma^\sigma(e) \downarrow = \varphi_e(e) \downarrow\}$$

is c.e. and has no prefix of G . Choose a prefix σ_0 of G with no extension in W . Define $h(e)$ by searching for any extension $\tau \succeq \sigma_0$ with $\Gamma^\tau(e) \downarrow$ and returning the value. Since Γ^G is total, this is a total computable function. The choice of σ_0 guarantees that h is DNR. But if j is a computable index for h , then $\varphi_j(j) = h(j)$, a contradiction. $\square$

Corollary 5.4. *Every non-high 1-generic has CA. In particular, every low 1-generic has CA.*

Proof. By Lemma 5.3 neither alternative in Theorem 5.2 holds for a non-high 1-generic. A low oracle is not high, by strictness of the Turing jump. $\square$

For completeness, the low examples needed in the main theorem can be obtained explicitly relative to the halting set.

Proposition 5.5 (Low Δ_2^0 generic witnesses). *There is a 1-generic $G \leq_T \emptyset'$. Every 1-generic below $\emptyset'$ is low and noncomputable.*

Proof. Enumerate all c.e. sets of finite strings as (W_e) . Starting with the empty string, at stage e use $\emptyset'$ to decide whether the current string has an extension in W_e . If yes, find such an extension by ordinary enumeration and adopt it; if not, keep the current string. Append a zero to ensure growth. The union G is computable from $\emptyset'$ and meets or avoids each W_e .

For any 1-generic G , membership in G' can be decided from $G \oplus \emptyset'$. For the computation $\Phi_e^G(e)$, search along prefixes of G for either a convergence witness or a prefix having no extension on which the computation converges. The latter test is decidable in $\emptyset'$. Genericity guarantees one of these alternatives. Hence $G' \leq_T G \oplus \emptyset'$. If $G \leq_T \emptyset'$, it follows that $G' \equiv_T \emptyset'$. Finally, a computable G would be a predictor of its own bits from the all-zero oracle $G_{\mathbb{N}}$, violating Lemma 3.2 with $T = \mathbb{N}$. $\square$

The generalized-low calculation in this proof is standard; it is also recalled in the discussion surrounding Proposition 5.5 of Astor–Hirschfeldt–Jockusch.[3] For our application the important point is the separation of dependencies: the direct 2-generic route is elementary, whereas the sharper low-witness route uses Theorem 5.2 in addition.

6 No least representative, even under an arbitrary sparse budget

6.1 The local theorem

Theorem 6.1. *Let G be 1-generic with $\text{CA}(G)$. Let $A \leq_T G$ and suppose some $d \in \text{ED}(G)$ is computable from A . For any unbounded computable order b , there is an infinite computable mask T satisfying all conclusions of Theorem 1.2.*

Proof. Let $S = S_d$. Then $S \in \mathcal{Z}_0$ and $S \leq_T A \leq_T G$. By Theorem 4.4, choose an infinite computable null set T obeying the budget and with $T \setminus S$ infinite. If $A \leq_T G_T$, then $d \leq_T G_T$, so Corollary 3.3 would imply $T \subseteq^* S$. This contradiction proves $A \not\leq_T G_T$.

Lemma 3.1 gives $G_T \equiv_{\text{ued}} G$ and $G_T \leq_T G$. The inequality is strict by Theorem 3.4, since T is infinite. Finally $G \triangle G_T \subseteq T$, so the same budget bounds the actual disagreements. $\square$

6.2 Checking the quantifiers of the original question

Theorem 6.2 (Negative answer to C2). *If G is 1-generic and has $\text{CA}(G)$, neither $[G]_{\text{ned}}$ nor $[G]_{\text{ued}}$ has an element of least Turing degree, even if only binary representatives are allowed.*

Proof. Take any proposed representative $g \in [G]_{\text{ned}}$. If $g \not\leq_T G$, the representative G already witnesses that g is not least. In fact every $G_T \leq_T G$ also fails to compute g .

Otherwise $g \leq_T G$. Since $G \leq_{\text{ned}} g$, implication (4) supplies $d \in \text{ED}(G)$ with $d \leq_T g$. Apply Theorem 6.1 with $A = g$. The resulting binary representative $G_T \equiv_{\text{ued}} G$ does not compute g . It belongs to the nonuniform class as well, so g is not least there.

If g was proposed as a least member only of $[G]_{\text{ued}}$, the same argument applies: this class is contained in $[G]_{\text{ned}}$, and both comparison witnesses, G and G_T , are in the uniform class. They are binary, so the proof also excludes a least degree in either binary subclass. $\square$

Together with Corollary 5.4 and Proposition 5.5, this proves Theorem 1.1, including the low Δ_2^0 refinement.

Corollary 6.3 (A countable test family already has no in-class lower bound). *Fix G as above and an unbounded computable order b . Let*

$$\mathcal{M}_b(G) = \{G_T : T \text{ computable, } T \in \mathcal{Z}_0, (\forall N) |T \cap [0, N]| \leq b(N)\}.$$

No member of $[G]_{\text{ned}}$ is a Turing lower bound of $\mathcal{M}_b(G)$. In particular the nonuniform class has no least member even when leastness is tested only against this countable family of binary representatives in the uniform class.

Proof. A lower bound g is below G , since the empty mask is allowed. As above, g computes a description of G . Theorem 6.1 supplies a member of $\mathcal{M}_b(G)$ not computing g . $\square$

6.3 Finite simultaneous avoidance and a sharp budget boundary

Corollary 6.4 (Finite targets). *For every finite family $g_0, \dots, g_{k-1} \in [G]_{\text{ned}}$ and every unbounded computable order b , one computable null mask T within budget satisfies $g_i \not\leq_T G_T$ for all $i < k$. Consequently neither the uniform nor the nonuniform class has a finite coinital subset.*

Proof. Any $g_i \not\leq_T G$ is automatically not below G_T . For each remaining i , choose $d_i \in \text{ED}(G)$ with $d_i \leq_T g_i \leq_T G$. The finite union $S = \bigcup_i S_{d_i}$ is null and computable from G . Use

Theorem 4.4 to make $T \setminus S$ infinite. Then $T \setminus S_{d_i}$ is infinite for each relevant i , and Corollary 3.3 rules out $g_i \leq_T G_T$. If there are no relevant targets, any infinite budgeted selector suffices. An equivalent representative not computing any member of the proposed finite family disproves coinitality. $\square$

Proposition 6.5 (Why unboundedness is necessary). *If $|T \cap [0, N]| \leq M$ for all N , then T is finite and $G_T \equiv_T G$. Such masks cannot witness $A \not\leq_T G_T$ for any $A \leq_T G$.*

Proof. An infinite T would have a prefix containing more than M elements. Finite changes preserve ordinary Turing degree by hard-coding the changed bits. $\square$

Thus the distinction between bounded and unbounded budgets is exact for this local erasure method. A fixed-budget neighborhood is not being called an equivalence class: budgets need not be preserved under composition.

7 Uniformly presented countable families

7.1 An effective version of the null-set P-ideal calculation

A countable family of null sets has a null upper bound modulo finite sets. For the present application the computational presentation of the family matters. The following direct construction needs a uniform membership procedure, but needs no density-convergence moduli and no extra jump.

Lemma 7.1 (Uniform null upper bounds). *Suppose $(S_i)_{i \in \mathbb{N}}$ is uniformly computable from an oracle G and each S_i is in $\mathcal{Z}_0$. There is $S \leq_T G$ in $\mathcal{Z}_0$ with $S_i \subseteq^* S$ for every i .*

Proof. Use the dyadic blocks $J_n = [2^n, 2^{n+1})$. For $k \geq 1$, define

$$r_k(n) = \frac{|\bigcup_{i < k} S_i \cap J_n|}{2^n}.$$

For each fixed k , $r_k(n) \rightarrow 0$, since the numerator is at most $|\bigcup_{i < k} S_i \cap [0, 2^{n+1})|$. Let

$$k(n) = \max(\{0\} \cup \{k : 1 \leq k \leq n+1 \text{ and } r_k(n) \leq 1/k\}).$$

All tests here are finite rational comparisons, uniformly decidable in G . For each fixed k , that k is eligible at all sufficiently large n . Hence $k(n) \rightarrow \infty$. Set

$$S \cap J_n = \left(\bigcup_{i < k(n)} S_i \right) \cap J_n, \quad 0 \notin S.$$

This defines $S \leq_T G$. Since $k(n) > i$ eventually, $S_i \subseteq^* S$. The fraction of S in block J_n is at most $1/k(n)$ when $k(n) > 0$, and is zero otherwise; it therefore tends to zero.

To pass from block fractions to full density, fix $\varepsilon > 0$ and choose n_0 so that all later block fractions are at most ε . For $2^n \leq N < 2^{n+1}$ with $n \geq n_0$,

$$|S \cap [0, N]| \leq 2^{n_0} + \varepsilon \sum_{j=n_0}^n 2^j \leq 2^{n_0} + \varepsilon 2^{n+1}.$$

Thus $\limsup_N \rho_N(S) \leq 2\varepsilon$. As ε was arbitrary, $S \in \mathcal{Z}_0$. $\square$

7.2 One computable mask defeating countably many oracles

Theorem 7.2 (Uniform countable simultaneous avoidance). *Let G be 1-generic with $\text{CA}(G)$. Suppose $(d_i)_{i \in \mathbb{N}}$ is a uniformly G -computable sequence, each $d_i \in \text{ED}(G)$. Let A_i be any oracles with $d_i \leq_T A_i$ for every i ; the latter reductions need not be uniform. For every unbounded computable order b , there is an infinite computable null mask T within budget such that*

$$(\forall i) \quad A_i \not\leq_T G_T.$$

Proof. The omission sets $S_i = S_{d_i}$ are uniformly G -computable and null. Lemma 7.1 supplies a null $S \leq_T G$ with $S_i \subseteq^* S$ for every i . By Theorem 4.4, choose a computable budgeted T with $T \setminus S$ infinite. For each i , only finitely many of these escaping points can belong to $S_i \setminus S$, so $T \setminus S_i$ is infinite. If $A_i \leq_T G_T$, then $d_i \leq_T G_T$, contradicting Corollary 3.3. $\square$

An immediate special case is that no uniformly G -computable family of effective-dense descriptions has the property that every computable-null erasure of G computes one of those descriptions. Theorem 7.2 also excludes a countable cointial family of representatives whenever suitable descriptions below its members have such a uniform presentation.

Remark 7.3 (The quantifier that cannot be discarded). A sequence of sets individually computable from G need not be uniformly computable from G . Likewise, the assertion that each proposed representative computes some description does not uniformly select indices for correct total descriptions. Lemma 7.1 makes finitely many queries across the first k rows on each block; it needs the uniform procedure for those rows. This paper therefore does not infer the absence of arbitrary countable cointial families from Theorem 7.2.

8 A packed descending chain with no in-class lower bound

8.1 A nonuniform enumeration is used openly

Fix an unbounded computable order b . Choose blocks $I_n = [m_n, 2m_n)$ from Lemma 4.1 with

$$a_n = 2(n+1), \quad Q_n = (n+1)(n+2).$$

Let $(h_i)_{i \in \mathbb{N}}$ list all total computable functions. This is a set-theoretic listing, *not* a computable enumeration of total computable functions. For instance, list increasingly the indices in TOT and take their computed functions. Membership in TOT is decidable in $\emptyset''$, so such a listing can be chosen $\emptyset''$ -computable at the level of indices. For each fixed i , the finite list $h_0, \dots, h_i$ consists of ordinary computable functions whose indices may be hard-coded.

Define for all i, n

$$p_i(n) = h_i(n) \bmod m_n.$$

For $n \geq i$, define recursively in i

$$q_i(n) = \min([0, m_n) \setminus (\{p_i(n)\} \cup \{p_j(n), q_j(n) : j < i\})). \quad (12)$$

At most $2i+1$ offsets are forbidden, while $m_n \geq 2n+3 > 2i+1$. Thus this value exists. For fixed i , it is computable on $n \geq i$. Values on smaller n are unused and may be set to zero.

Let C_i be empty outside the blocks and set

$$C_i \cap I_n = \{m_n + p_j(n), m_n + q_j(n) : 0 \leq j \leq \min(i, n)\}. \quad (13)$$

Each C_i is computable. To decide membership, locate the relevant block and run only the finitely many computable functions with indices at most i . There is no assertion that $i \mapsto$ an ordinary computable index for C_i is itself computable.

8.2 The geometry and strict descent

Lemma 8.1. *The masks satisfy $C_i \subseteq C_{i+1}$ and $|C_{i+1} \setminus C_i| = \infty$. Their union $C_\infty = \bigcup_i C_i$ is null and obeys the budget b at every prefix.*

Proof. Inclusion is immediate from (13). For each $n \geq i + 1$, the point $m_n + q_{i+1}(n)$ belongs to C_{i+1} but not to C_i , by its freshness requirement. These points are distinct in different blocks. There are at most $2(n + 1)$ points of C_∞ in block I_n , so Lemma 4.1 gives both its nullity and the budget. Every C_i inherits them. The union need not be computable. $\square$

Theorem 8.2 (Descending chain theorem). *Let G be 1-generic with $\text{CA}(G)$. For the masks above, $G_i = G_{C_i}$ satisfies all conclusions of Theorem 1.3.*

Proof. Lemma 3.1 gives $G_i \equiv_{\text{ued}} G$ and $G_i \leq_T G$. Theorem 3.4 and Lemma 8.1 give $G_{i+1} <_T G_i$. The budget bounds actual disagreements because $G \triangle G_i \subseteq C_i$.

Suppose, toward a contradiction, that $g \in [G]_{\text{ned}}$ and $g \leq_T G_i$ for every i . Then $g \leq_T G$. Choose $d \in \text{ED}(G)$ with $d \leq_T g$, and let $S = S_d$. We have $S \leq_T G$ and $S \in \mathcal{Z}_0$. For every i , $d \leq_T G_{C_i}$, so Corollary 3.3 gives

$$C_i \subseteq^* S. \quad (14)$$

For the particular total computable function h_i , the selector T_{h_i} from (9) has all of its points on blocks $n \geq i$ in C_i . Thus (14) implies $T_{h_i} \subseteq^* S$. Every total computable function appears in the list, so Lemma 4.3 makes $f_S \leq_T G$ eventually different from every computable function. This contradicts $\text{CA}(G)$.

If G is low, monotonicity of the jump yields $\emptyset' \leq_T G'_i \leq_T G' \equiv_T \emptyset'$. Hence each G_i is low. $\square$

All these degrees are noncomputable. For example, Theorem 3.4, with $D = \mathbb{N}$, would make $G_{C_i} \leq_T G_{\mathbb{N}}$ imply $\mathbb{N} \subseteq^* C_i$, impossible for a null mask. Thus, choosing the low generic from Proposition 5.5, the construction gives an infinite strictly descending chain of nonzero low Turing degrees in one uniform effective-dense class, without an in-class lower bound even in the larger nonuniform class.

8.3 Why the countable union does not give a counterargument

The union C_∞ in Lemma 8.1 really is null; the argument does not exploit a failure of closure under arbitrary unions. But $C_\infty \not\leq_T G$. Otherwise its containment of each selector tail would, by Lemma 4.3, again give an eventually different function below G . Consequently one cannot apply the computable-mask transformation to C_∞ .

There is no contradiction between this construction and Lemma 7.1. The individual masks are computable, but the chosen listing of them is not asserted to be uniformly G -computable. The extra fresh points in (12) are also important: selector functions can coincide infinitely often, so merely adjoining the next selector need not make each consecutive degree drop strict.

For a low $G \leq_T \emptyset'$, the entire displayed family of masks and masked sets has a uniform $\emptyset''$ presentation using the index listing through TOT. This is an upper bound on the presentation, not a claim that the bound is optimal.

9 Finite degree patterns and interpretation of the obstruction

9.1 Boolean intervals inside an arbitrarily small neighborhood

Corollary 9.1 (Finite Boolean patterns). *Let G be 1-generic, let b be an unbounded computable order, and let $k \geq 1$. Inside $[G]_{\text{ued}}$ there are 2^k distinct ordinary degrees forming a Boolean*

lattice under the actual Turing joins and meets. All representatives can be chosen below G and within disagreement budget b of G . If G is low, all these degrees are low.

Proof. Choose an infinite computable null set T within budget using one point per block. Split its computable increasing enumeration into k infinite computable pieces $T_0, \dots, T_{k-1}$ by the enumeration index modulo k . For $u \subseteq \{0, \dots, k-1\}$, let $C_u = \bigcup_{i \in u} T_i$. All masks are subsets of T . Distinct u give masks different on an infinite set, so their degrees are distinct by Theorem 3.4. That theorem reverses inclusion, and Theorem 3.5 identifies union with meet and intersection with join. This is the dual Boolean lattice, which is itself Boolean. The low assertion follows by jump monotonicity as before. $\square$

For $k = 2$, the two intermediate degrees are incomparable, their join is $\deg_T(G)$, and their meet is $\deg_T(G_T)$. The meet is nonzero because T is null. Thus these are not ordinary minimal pairs. The finite geometry is compatible with the complete lack of a least degree in the full class under the stronger hypotheses of Theorem 6.2.

9.2 The repository gap, resolved at the required quantifier level

The plan's C2 argument would make the omission set S of a least representative almost contain every computable null set. The proposed contradiction there was not a purely set-theoretic one: null sets form a P-ideal, so abstract upper bounds exist.[1] The present proof locates the missing computational obstruction:

$$\begin{aligned} S \leq_T G, \quad S \in \mathcal{Z}_0, \quad (\forall T \text{ computable null}) \quad T \subseteq^* S \\ \implies \quad G \text{ computes an eventually different function.} \end{aligned}$$

For 2-generics, or non-high 1-generics, that consequence is impossible. The general least-representative assertion is universal in the starting function, so this subclass is already sufficient for a negative answer. We have not silently completed the original sketch for every 1-generic.

[write] In the plan's terms (`turing_degrees_unified.tex`, line 570): step (i) is Lemma 3.1, with the value 0 instead of 2 on the mask; step (ii) is Corollary 3.3; step (iii) is replaced by the hypothesis $CA(G)$, which fails exactly for the high 1-generics. The title of this subsection should therefore be read as “bypassed for 1-generics with CA ”: the plan's gap is re-scoped to the high 1-generics of Question 10.1, not closed. The C2 question itself, being universal, is answered negatively.

Two logically distinct facts do the work. Genericity forbids a retained oracle from making infinitely many error-free predictions on deleted coordinates. Computable agreement prevents a G -computable null omission set from swallowing all computable sparse selectors. Neither fact alone is the least-representative theorem.

9.3 Six boundaries of the conclusion

The following distinctions prevent overstatements of what has been proved.

Descriptions versus representatives. A description is allowed to contain $\perp$; a representative is a numerical function in the equivalence class. The proof extracts a description from a representative by (4). It never identifies the entire representative spectrum with an upward closure of description degrees.

Explicit omission versus an erroneous answer. The erased sets use numerical zero, but the reverse description reduction *explicitly* replaces every masked coordinate by $\perp$. No wrong answer is reclassified as an omission after the fact.

Least versus minimal. The mask family has no minimal element, and the full class has no least element. The existence of a minimal element of the *full* class is not settled by these results. A representative outside the displayed family need not have a smaller displayed representative below it.

Finite versus countable uniformity. Finite unions of omission sets are G -computable when the constituent sets are. Countable unions do not enjoy that inference without a uniform presentation and a null-upper-bound construction. Section 7 states exactly the uniform hypothesis it uses.

Actual degree order versus asymptotic degree order. The strict inequalities in the chain are ordinary Turing inequalities. All chain members are equivalent for uniform effective-dense reducibility. There is no strict chain of effective-dense degrees in that assertion.

Provability versus priority and formal certification. The arguments constitute a conventional proof submission. They are not results certified by finite experiments, by repository inclusion, by peer review, or by a proof assistant. The low sharpening has the named classical input of Theorem 5.2; it is not an unproved assumption introduced to obtain the conclusion.

10 Further research questions and concrete next targets

Except where linked to a cited question, the following are questions raised by this development, not claims that a comprehensive search has established their open status. Their formulations separate plausible next lemmas from large classification problems.

10.1 Genericity and computational strength

Question 10.1 (The high 1-generic case). Does $[G]_{\text{ned}}$ have no least Turing degree for every high 1-generic G ? Does the same hold for $[G]_{\text{ued}}$?

The present argument handles all non-high 1-generics. In the remaining case $\text{CA}(G)$ fails by Theorem 5.2. That failure only removes our obstruction; it does not construct a least representative. A possible next step is to use that a putative omission set is computable from *every* computable-null erasure, a stronger demand than just $S \leq_T G$.

Question 10.2 (Computational strength of null upper bounds). Which degrees compute an $S \in \mathcal{Z}_0$ such that $T \subseteq^* S$ for every computable $T \in \mathcal{Z}_0$? Does the answer change if T is restricted to one-point-per-block selectors for a fixed computable block layout?

Lemma 4.3 gives a necessary condition: such a degree computes a computably bounded eventually different function, and hence is high or DNR by Theorem 5.2. Sufficiency, the role of a fixed bound, and the difference between all null sets and selectors should be investigated separately. A proof for selectors alone would not automatically solve the full null-cover question.

Question 10.3 (How low can the starting object be?). Which nonzero degrees bound a binary set whose effective-dense class has no least ordinary degree? Is there a c.e. binary counterexample? What effective bounds can be imposed on descriptions witnessing its nontriviality?

Low Δ_2^0 examples now exist by Theorem 6.2, so merely asking for an arithmetical or halting-computable counterexample is no longer a useful target here. The c.e. restriction is a different structural demand.

10.2 Spectra and countable lower structure

Question 10.4 (Minimal representatives). Does either full class of a low 1-generic contain a representative of minimal ordinary Turing degree within that class?

The mask lattice and packed chain show extensive descent, but not descent below an arbitrary representative. A promising restricted goal is to characterize when a representative below G has an effective-dense equivalent strictly below it. Such a theorem would have to construct a new representative below that prescribed oracle, rather than only one that fails to compute it.

Question 10.5 (Arbitrary countable coinitality). Can $[G]_{\text{ned}}$ or $[G]_{\text{ued}}$ have a countable coinital family when G is a low 1-generic? Can the uniformity hypothesis of Theorem 7.2 be removed or replaced by a weaker effective presentation condition?

The packed chain has no lower bound in the class, but it need not be coinital. Theorem 7.2 eliminates uniformly supplied descriptions; it does not select correct descriptions from an arbitrary countable list of representatives. Pinning down that selection complexity is a concrete intermediate target.

Question 10.6 (The common lower cone of all computable erasures). For a 1-generic G , determine

$$\bigcap_{T \in \text{Comp} \cap \mathcal{Z}_0} \text{Low}(G_T).$$

When G is low, is this ideal trivial? Can it be nonprincipal?

For two masks, Theorem 3.5 gives an exact answer. For the full intersection, the present paper only excludes members that compute an effective-dense description of G , under $\text{CA}(G)$. A common lower bound may carry information without itself providing any such description.

Question 10.7 (Classifying classes with least degrees). Find intrinsic necessary and sufficient conditions for an effective-dense class to have a least ordinary Turing degree, separately for the uniform and nonuniform versions.

The criterion should distinguish genuine recoverability from every *description* from lower-boundedness among *representatives*. The repository's positive examples based on redundantly coded dyadic columns provide a comparison family, but its coarse characterization cannot simply be reused.[1]

10.3 Quantitative, effective, and formal directions

Question 10.8 (Effective presentations of descending chains). What is the least possible complexity of a sequence of computable masks whose erased oracles form a strict chain with no lower bound in the class? Can a uniformly G -computable presentation occur without yielding a uniformly G -computable sequence of descriptions?

Section 8 gives a $\emptyset''$ presentation for the low witnesses and carefully separates it from individual computability. Determining the optimal bound requires an argument about presentations, not just the degree of each separate mask.

Question 10.9 (Other small-set ideals). For which effectively presented ideals $\mathcal{I}$ of negligible sets can the sparse escape theorem and no-least argument be repeated, with $\mathcal{Z}_0$ replaced by $\mathcal{I}$?

The proof suggests explicit hypotheses: computable masks should allow the two description transformations; an $\mathcal{I}$ -small omission set should miss a point in every sufficiently late block of a chosen layout; and computable selectors should lie in $\mathcal{I}$ with the desired budget. Weighted-density and summable ideals are natural tests, but these hypotheses must be checked rather than assumed.

Question 10.10 (Randomness instead of genericity). Which randomness hypotheses give an analogue of Lemma 3.2 for computable coordinate erasures, and which of those oracles also admit a suitable sparse escape theorem?

This separates a nonprediction problem from a highness problem. Quasiminimality results for random sets in asymptotic degree structures do not on their own settle this representative-spectrum question.[3]

Question 10.11 (Binary representative complexity). Can the description-selection machinery here contribute to a bound on the complexity of a binary representative of a numerical function already known to be nonuniformly effective-dense equivalent to some set?

This is a possible connection to Gerdes’s Question 5 and repository C3, not a solution of either. The present counterexamples start with a binary set, so they avoid rather than solve that selection problem.[2, 1]

Question 10.12 (Formalization of the proof interface). Can generic nonprediction, budgeted sparse escape, and the class-level leastness argument be formalized as reusable modules over a common oracle computation interface in the repository?

A useful first milestone is a checked theorem taking 1-genericity and $\text{CA}(G)$ as explicit hypotheses and proving Theorem 6.2. The genericity constructions and the classical highness refinement should then be proved or imported with their dependencies audited separately.

11 Proof audit, formalization plan, and reproducibility

11.1 The dependency structure

The shortest independent route to C2 has the following structure:

$$\begin{aligned} & 2\text{-genericity} \implies \text{CA}(G) \\ & \text{CA}(G) + \text{null omission set} \implies \text{computable sparse escape} \\ & 1\text{-genericity} + \text{description below } G_T \implies T \subseteq^* S_d \\ & \text{least representative} \implies \text{contradiction.} \end{aligned}$$

A 2-generic exists below $\emptyset''$: enumerate the Σ_2^0 sets of strings and, using $\emptyset''$ to decide whether an extension exists, meet the current set when possible and otherwise preserve an avoiding prefix. Append a bit at each stage. This is the same meet-or-avoid construction as in Proposition 5.5, one jump higher. Hence the direct route has actual witnesses without appealing to Theorem 5.2.

The low route instead uses

$$\text{low 1-generic} \implies \text{not high and not DNR} \xrightarrow{\text{classical Theorem 5.2}} \text{CA}(G).$$

The only named external classification in this route is stated in full at its point of use. Neither route imports the repository’s unrefereed coarse-spectrum theorems as black boxes.

11.2 A practical formalization boundary

An end-to-end formalization needs more than the quotient definition of Turing degrees. The essential interfaces are finite-use oracle computations with extension persistence; c.e. and Σ_2^0 predicates on strings; explicit meet-or-avoid genericity; coding of total descriptions with a distinguished omission symbol; and a density-zero library with finite unions and prefix-count bounds.

The central theorem can first be proved for sets and total function oracles, postponing passage to the degree quotient. Generic nonprediction is a finite-computation argument. Budgeted blocks, selectors, and the uniform P-ideal calculation are arithmetic constructions with eventual bounds. The description-equivalence maps have direct pointwise proofs. Only after these components are available does the class-level theorem become a short composition of implications.

Theorem 5.1 additionally needs the complexity calculation for B_Γ and the c.e. dense sets U_k . The low refinement needs jump monotonicity, the generalized-low computation, and the classical high-or-DNR characterization. Importing the latter as an axiom would make the low result conditional in the formal development, even though it is a published theorem in the mathematical argument. No Lean code in this archive is represented as compiled or certified.

11.3 What the accompanying finite checks do

The companion script `checks/check_finite.py` is deterministic and uses only the Python standard library. It checks the two pointwise correctness transformations for all small binary strings and masks; finite erasure independence under a changed deleted bit; the join and finite-amalgamation coordinate rules; capacity and freshness invariants of the packed masks; and every-prefix budgets for representative finite block layouts. It also checks finite instances of the dyadic-block inequality used in Lemma 7.1.

These checks are aimed at off-by-one, coordinate, and counting errors. They do not test that a finite string is generic, exhibit a noncomputable oracle in finite time, decide which programs are total, or verify an infinite density limit. The proof of each such assertion is the corresponding conventional argument above. The recorded output and a machine-readable summary are supplied separately; rerunning the script regenerates them.

The document is built by `build.sh`, which runs the checks and compiles `article.tex` with pdfLaTeX. The archive contains the source, final PDF, build instructions, source ledger, and proof audit. Fonts are supplied by the user’s TeX installation, not bundled in the archive.

[write] In report 14 the script is `code/check_finite.py`; its recorded outputs are `data/results.json` and `data/run.txt` (byte-identical: the script prints the same text it writes). The script writes `results.json` beside itself, and the delivered `build.sh`, shipped as `code/build.sh`, changes to its own directory, writes `checks/run.txt` there, compiles `article.tex` and copies the PDF over `article.pdf`; neither should be run in place. The delivered PDF is not shipped; the report’s `README.md` gives working commands on a copy.

A Source provenance and novelty audit

A.1 Repository snapshot

The task was anchored to PROVEIT commit

1a1396d4d3a2ac6812692df520517d86ba3a4785.

The primary repository document is

Computability/TuringDegrees/Research/CoarseDegrees/research-plan/turing_degrees_unified.tex.

The fetched blob is `ebecd2e15f8e4a06ebd0b0333387d27ebe630375`. In that snapshot, source lines 552–573 include the established coarse tools, the C2 question, its genericity sketch, and the explicit gap. The C3–C8 follow-ups begin in the immediately following range. The source describes itself as the second edition of a research synthesis; it is not treated here as an externally refereed source.

A.2 Primary literature checked

Gerdes’s arXiv record displayed version 1, submitted 9 August 2025, when consulted on 28 September 2026. Section 7, Question 7 is the precise least-representative question targeted here. The nearby questions about binary representatives and descriptive complexity concern different problems.[2]

The Astor–Hirschfeldt–Jockusch preprint was checked for the definition of effective-dense descriptions, the uniform and nonuniform reductions, and the distinction between quasiminimality and internal ordinary-degree structure. Its Theorem 5.9 is not cited as an internal least-degree theorem. The paper’s minimal-pair results also concern distinct questions about asymptotic degrees.[3]

The Brendle–Brooke–Taylor–Ng–Nies paper records the equivalence between computing an eventually different function and weak meager engulfing, and attributes the exclusion of 2-generics from that class to Rupprecht. The direct proof in Section 5 is included so that the main application does not depend on translating that terminology.[5]

The high-or-DNR characterization was checked directly in Kjos-Hanssen–Merkle–Stephan, Theorem 5.1, rather than inferred only from a secondary reference. It is used solely for the non-high and low refinements.[4]

A.3 What the audit does not establish

Targeted searches for the combination of effective-dense representatives, least ordinary Turing degrees, genericity, and eventually different functions did not establish a prior occurrence of the exact erasure application proved here. This is a limited search result, not evidence that no earlier proof exists. In particular, elementary applications of standard genericity may circulate in unpublished notes or use different terminology. The mathematical claim is the stated theorem with its proof; priority remains to be assessed independently.

B A compact checklist of delicate proof obligations

Issue	Where and how it is handled
Wrong values versus omissions	Lemma 3.1 gives two explicit transformations. The inverse discards the whole computable mask.
Full-class leastness	Theorem 6.2 treats numerical representatives, including those not below G , and uses binary comparison witnesses in the uniform class.
Nonuniform hard-coded data	The generic avoiding prefix, the computable agreement index, and each finite list of total-function indices are allowed to depend on the instance. No uniform selector is asserted.

Issue	Where and how it is handled
Decidable sparse mask	Equation (9) gives one point in each decidable interval using a total computable function, not a delayed c.e. enumeration.
Density at intermediate prefixes	Lemma 4.1 treats every $N \in [m_n, m_{n+1})$; Lemma 7.1 gives an explicit partial-block bound.
Bounded default in f_S	Density zero makes the default branch in (8) occur only finitely often. Infinite agreement remains infinite after those exceptions are discarded.
Infinite strict drops	The offsets $q_i(n)$ are fresh from all previously inserted points; coincident selector functions cannot collapse a step.
Countable presentations	Section 7 requires a uniform G procedure. Section 8 uses an explicitly nonuniform list of total computable functions.
Actual meet in the full degree order	The amalgamation argument of Theorem 3.5 handles any common lower oracle, not just another computable-mask oracle.
Low refinement	Theorem 5.2 is a named published input. Generic non-DNR and generalized lowness are proved separately.
No overclaim about minimality	The full-class theorem excludes least elements; Question 10.4 keeps minimal elements separate.
No experimental certification	Finite checks validate finite invariants only. No independent review or proof-assistant verification is claimed.

References

- [1] PROVEIT repository, *Turing Degrees: A Unified Research Report*, prepared for Vladimir Reshetnikov, research-plan source, second edition dated 18 September 2026, repository snapshot 1a1396d4d3a2ac6812692df520517d86ba3a4785. https://github.com/VladimirReshetnikov/ProveIt/blob/1a1396d4d3a2ac6812692df520517d86ba3a4785/Computability/TuringDegrees/Research/CoarseDegrees/research-plan/turing_degrees_unified.tex. Consulted 28 September 2026. Unrefereed research programme.
- [2] Peter M. Gerdes, *Comparing Notions of Dense Computability on ω^ω and 2^ω* , arXiv:2508.06925v1, 2025. <https://arxiv.org/abs/2508.06925v1>. In particular, Section 7, Questions 5 and 7.
- [3] Eric P. Astor, Denis R. Hirschfeldt, and Carl G. Jockusch, Jr., *Dense computability, upper cones, and minimal pairs*, *Computability* **8** (2019), 155–177. DOI: 10.3233/COM-180231. Author-hosted preprint: <https://www.math.uchicago.edu/~drh/Papers/Papers/dense.pdf>.
- [4] Bjørn Kjos-Hanssen, Wolfgang Merkle, and Frank Stephan, *Kolmogorov complexity and the Recursion Theorem*, *Transactions of the American Mathematical Society* **363** (2011), 5465–5480. Preprint arXiv:0901.3933v2. <https://arxiv.org/abs/0901.3933v2>. Theorem 5.1 is the high-or-DNR characterization used here.
- [5] Jörg Brendle, Andrew Brooke-Taylor, Keng Meng Ng, and André Nies, *An analogy between cardinal characteristics and highness properties of oracles*, arXiv:1404.2839v2, 2014. <https://arxiv.org/abs/1404.2839v2>. See Section 3.2, Theorem 4.2, and the discussion of 2-genericity in Section 4.